The coupled field equations of the 7+1 model and their solutions in the three gravity regimes

Every interaction in equation form; accurate approximate solutions in the primordial regime (cubic, Gauss–Bonnet and Einstein sub-regimes), the modern regime and the far future, checked against high-accuracy numerical solutions — sections 20–23 of the notebook 7+1.nb

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Abstract

The 7+1 model has gravity with all three Lovelock terms, the 16-component field dirac16complex with its mass, its nine self-couplings and its spin-connection coupling to gravity, a stabilizing potential of the auxiliary space, and the Standard Model on a world brane with three portals to dirac16complex. This document writes every one of these interactions as an equation, derives the coupled field equations and their homogeneous cosmological form, and solves them in the three gravity regimes: (I) the PRIMORDIAL regime near $x_0 = 0$, where the cubic, then the Gauss–Bonnet, then the Einstein term dominates; (II) the MODERN regime, where the auxiliary space is held by its potential and gravity on the brane is four-dimensional; (III) the FAR FUTURE, de Sitter expansion. Each regime has closed-form solutions (exact where they exist) whose accuracy is measured against the numerical solution of the full equations. Sections 20–23 of 7+1.nb contain 48 checks (48 True); the independent Python program tools/ref_coupled.py contains 50 (50 True); the 34 groups of numbers printed by both agree to a largest relative difference of 0.00032.

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1. Results at a glance

- **Every interaction** of the model is a term of one action, $S = S_0 + S_1 + S_2 + S_3 + S_\psi + S_m + S_{\text{self}} + S_U + S_f + S_{\text{SM}} + S_P$ (section 3). The cubic Lovelock density in invariant form is $L_3 = R^3 - 12RR_{\mu\nu}R^{\mu\nu} + 16R^\mu{}_\nu R^\nu{}_\rho R^\rho{}_\mu + 24R^{\mu\rho}R^{\nu\sigma}R_{\mu\nu\rho\sigma} + 3RR_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 24R^{\mu\nu}R_\mu{}^{\rho\sigma\tau}R_{\nu\rho\sigma\tau} + 4R^{\mu\nu}{}_{\rho\sigma}R^{\rho\sigma}{}_{\lambda\tau}R^{\lambda\tau}{}_{\mu\nu} - 8R_\mu{}^\rho{}_\nu{}^\sigma R_\rho{}^\lambda{}_\sigma{}^\tau R_\lambda{}^\mu{}_\tau{}^\nu$ (checked exactly).
- **The radion** $\beta = \ln(b/b_0)$ obeys $\beta'' + 3H_a\beta' + 4\beta'^2 = \kappa(\varepsilon/6 - p_a/2 + p_b/3) + \Lambda/3$; with $U = U_0 + \frac{K_U}{2}\beta^2$ its mass is $m_r^2 = \kappa K_U/12$; on the brane $8\pi G_4 = \kappa/V_\perp$, $\Lambda_4 = \Lambda + \kappa U_0 + \kappa f_W/V_\perp$ (section 5).
- **Regime I, vacuum: exact solution.** With $u = H_b/H_a$ the constraint is $P_E(u) + H_a^2 P_G(u) + \xi H_a^4 P_C(u) = 0$ — quadratic in H_a^2 — so the vacuum solution is an explicit curve plus one quadrature. Its limits are the cubic regime ($a \propto \tau$, $H_b \to -\sqrt{\alpha_2/(30\alpha_3)}$), the Gauss–Bonnet Kasner regime ($p_2 = 1.56115$, $q_2 = -0.420861$) and the Einstein Kasner regime ($p_1 = 0.546918$, $q_1 = -0.160189$); their asymptotic formulas are accurate to 6.7×10^{-6}, 0.00085 and 0.00032 inside the regimes (section 6).
- **Regime I with dirac16complex:** near $x_0 = 0$ the gas wins: $a \propto \tau^{3/4}$, $H_b \to h_* = -\sqrt{\alpha_2/(54\alpha_3)}$ (for any fluid $p_a = w_a\varepsilon$: $a \propto \tau^{1/(1+w_a)}$); later the gas dominates the Einstein regime and, WITHOUT a stabilizing potential, the auxiliary space re-expands, $b \propto \tau^{2/5}$ (section 6.4).
- **Regime II:** the potential $U(b)$ catches the auxiliary space; the radion then oscillates, $\beta \approx A\sin(m_r\tau + \vartheta)/(m_r(\tau - \tau_E))$, and carries 0.991 of the energy — the cosmological moduli problem of the minimal model: a 10 GeV radion decays in 0.11 s, but into the Kaluza–Klein modes of the bulk; through brane fields alone it would reheat the brane to only 1.0×10^{-9} GeV. After a later reheating, the brane obeys the Friedmann equations with $G_4 = G_8/V_\perp$; the composite analytic solution of the late universe is accurate to 0.00022 and gives the age 13.791 Gyr (section 7).
- **Regime III:** de Sitter with $H_\infty = H_0\Omega_\Lambda^{1/2}$, $H_a = H_\infty \coth(3H_\infty t/2)$; the radion is stable (friction γ, mass m_{eff} given in closed form with the Gauss–Bonnet and cubic terms) and settles at $\beta_{\text{eq}} = H^2(1 - 4\alpha_2 H^2)/m_r^2$ (section 8).
- **The size of every interaction** in the three regimes is tabulated in section 9.

2. Conventions and objects used

Units $\hbar = c = 1$; coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates. Greek indices $\mu, \nu, \cdots = 0, \ldots, 7$ are coordinate indices, capital Latin $A, B, \cdots = 0, \ldots, 7$ frame indices; repeated indices are summed. The frame metric is $\eta = \text{diag}(+1, -1, \ldots, -1)$, simply the squares of the octonion basis.

The signature: $\{7, 1, 0\}$ and $\text{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \text{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \text{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \text{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\text{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation

$Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

The real 16×16 Dirac matrices, the Dirac conjugate and the chirality:

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \quad \Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\eta_{AB} 1_{16},$$
$$\Gamma^A = \eta^{AA} \Gamma_A, \quad \bar\psi = \psi^\dagger \Gamma_0, \quad \chi = i\,\Gamma_0 \Gamma_1 \cdots \Gamma_7,$$

with real antisymmetric 8×8 matrices τ_i^O ($(\tau_i^O)^2 = -1_8$, mutually anticommuting). Curvature (Landau–Lifshitz):

$$\{{}^{\lambda}_{\mu\nu}\} = \tfrac{1}{2} g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}), \qquad R^\rho{}_{\sigma\mu\nu} = \partial_\mu \{{}^{\rho}_{\sigma\nu}\} - \partial_\nu \{{}^{\rho}_{\sigma\mu}\} + \{{}^{\rho}_{\mu\lambda}\}\{{}^{\lambda}_{\sigma\nu}\} - \{{}^{\rho}_{\nu\lambda}\}\{{}^{\lambda}_{\sigma\mu}\},$$
$$R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}, \qquad R = g^{\sigma\nu} R_{\sigma\nu}, \qquad G^\mu{}_\nu = R^\mu{}_\nu - \tfrac{1}{2}\delta^\mu_\nu R.$$

The generalized Kronecker delta $\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det[\delta^{\mu_i}_{\nu_j}]$ gives the Lovelock densities and tensors ($k = 0,1,2,3$; $L_0 = 1$, $E^{(0)\mu}{}_\nu = -\delta^\mu_\nu/2$, $L_1 = R$, $E^{(1)} = G$; $E^{(4)} \equiv 0$ in eight dimensions):

$$L_k = \frac{1}{2^k}\, \delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$
$$E^{(k)\mu}{}_\nu = -\frac{1}{2^{k+1}}\, \delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k}\, R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \qquad \delta\Big(\sqrt{|g|}\, L_k\Big) = \sqrt{|g|}\, E^{(k)}_{\mu\nu}\, \delta g^{\mu\nu}.$$

Frames, frame matrices and the spin connection (the coupling of dirac16complex to gravity): with $g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu$,

$$\Gamma^\mu(X) = e_A{}^\mu \Gamma^A, \qquad \omega_\mu{}^A{}_B = e^A{}_\nu \big(\partial_\mu e_B{}^\nu + \{{}^{\nu}_{\mu\lambda}\} e_B{}^\lambda\big),$$
$$\Omega_\mu = \tfrac{1}{4}\eta_{AC}\, \omega_\mu{}^C{}_B\, \Gamma^A \Gamma^B, \qquad D_\mu \psi = \partial_\mu \psi + \Omega_\mu \psi.$$

The homogeneous cosmological metric of the three regimes, its Hubble rates (a prime is d/dx_0) and the time since the singularity $\tau = x_0 - x_s$:

$$ds^2 = n^2 dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \qquad n = 1, \qquad H_a = \frac{a'}{a}, \qquad H_b = \frac{b'}{b}.$$

The couplings: $\kappa = 8\pi G_8 = M_8^{-6}$ (M_8 the reduced eight-dimensional Planck mass), $\alpha_1 = 1$, α_2 (Gauss–Bonnet, dimension length^2) and α_3 (cubic, length^4), $\alpha_0 = 2\Lambda$ with the bulk cosmological constant Λ. The Gauss–Bonnet length $\ell_2 = \sqrt{\alpha_2}$ and the ratio $\xi = \alpha_3/\alpha_2^2$ set the primordial regime. The auxiliary space is a torus T^4 of coordinate period L_c, proper volume $V_\perp = b^4 L_c^4$; the Standard Model lives on the world brane $W = \{x_4 = \cdots = x_7 = 0\}$ with induced metric $h_{\mu\nu} = g_{\mu\nu}(x, y = 0)$ ($y = (x_4, \ldots, x_7)$).

The three gravity regimes. The regimes are the three epochs of the model in which gravity behaves in three qualitatively different ways: (I) the primordial regime near $x_0 = 0$, governed by the higher Lovelock terms (inside it the cubic, the Gauss–Bonnet and the Einstein term take over one after another — three sub-regimes); (II) the modern regime, in which the auxiliary space is held by its stabilizing potential and gravity on the brane is four-dimensional Einstein gravity; (III) the far future, in which the four-dimensional cosmological constant dominates (de Sitter).

3. Every interaction in equation form (section 20.1)

The action of the model is the sum of the following terms ($\sqrt{|g|}\,d^8X$ is the volume element of the bulk, $\sqrt{|h|}\,d^4x$ that of the brane; a brane integral is $\int_W d^4x\sqrt{|h|}(\dots) = \int d^8X\sqrt{|g|}\,\delta^4(y)\,b^{-4}(\dots)$):

$$
\begin{aligned}
S_0 &= -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\,2\Lambda && \text{(cosmological term, } k=0\text{)},\\
S_1 &= -\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\,R && \text{(Einstein–Hilbert, } k=1\text{)},\\
S_2 &= -\frac{\alpha_2}{2\kappa}\int d^8X\sqrt{|g|}\,\big(R^2-4R_{\mu\nu}R^{\mu\nu}+R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\big) && \text{(Gauss–Bonnet)},\\
S_3 &= -\frac{\alpha_3}{2\kappa}\int d^8X\sqrt{|g|}\,L_3 && \text{(cubic Lovelock)},\\
S_\psi &= \int d^8X\sqrt{|g|}\;\bar\psi\,i\Gamma^\mu(X)\big(\partial_\mu+\Omega_\mu\big)\psi && \text{(dirac16complex, gravity)},\\
S_m &= -\int d^8X\sqrt{|g|}\;m\,\bar\psi\psi && \text{(mass)},\\
S_{\rm self} &= -\frac12\sum_{r=0}^{8} g_r\int d^8X\sqrt{|g|}\sum_{|K|=r}\eta_K\big(\bar\psi\Theta^K\psi\big)^2 && \text{(self-couplings)},\\
S_U &= -\int d^8X\sqrt{|g|}\;U(b) && \text{(stabilizing potential)},\\
S_f &= -\int_W d^4x\sqrt{|h|}\;f_W && \text{(brane tension)},\\
S_{\rm SM} &= \int_W d^4x\sqrt{|h|}\;\mathcal{L}_{\rm SM} && \text{(Standard Model)},\\
S_P &= \int_W d^4x\sqrt{|h|}\;\Big[-\sum_{i,s}y_{is}\,\bar L_i\tilde\Phi P_R\psi_s+\text{h.c.} && \text{(portals)}\\
&\qquad -G_P^f\,(\bar\psi\Gamma^\mu\psi)|_W(\bar f\gamma_\mu f)-\lambda_P\,\Phi^\dagger\Phi\,(\bar\psi\psi)|_W\Big].
\end{aligned}
$$

Here the cubic Lovelock density in invariant form is (checked EXACTLY in section 20.1 of 7+1.nb against the generalized-delta definition on four random integer tensors with all algebraic symmetries of the Riemann tensor, $R_{abcd}=S_{ac}S_{bd}-S_{ad}S_{bc}$ summed over three random symmetric integer matrices S):

$$
\begin{aligned}
L_3 &= R^3-12R\,R_{\mu\nu}R^{\mu\nu}+16R^\mu{}_\nu R^\nu{}_\rho R^\rho{}_\mu+24R^{\mu\rho}R^{\nu\sigma}R_{\mu\nu\rho\sigma}+3R\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\\
&\quad -24R^{\mu\nu}R_\mu{}^{\rho\sigma\tau}R_{\nu\rho\sigma\tau}+4R^{\mu\nu}{}_{\rho\sigma}R^{\rho\sigma}{}_{\lambda\tau}R^{\lambda\tau}{}_{\mu\nu}-8R_\mu{}^\rho{}_\nu{}^\sigma R_\rho{}^\lambda{}_\sigma{}^\tau R_\lambda{}^\mu{}_\tau{}^\nu\\
&= \frac18\,\delta^{\alpha_1\beta_1\alpha_2\beta_2\alpha_3\beta_3}_{\gamma_1\delta_1\gamma_2\delta_2\gamma_3\delta_3}R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}R^{\gamma_2\delta_2}{}_{\alpha_2\beta_2}R^{\gamma_3\delta_3}{}_{\alpha_3\beta_3}.
\end{aligned}
$$

The self-couplings use, for every strictly increasing multi-index $K=(A_1<\dots<A_r)$ of rank r (256 in all), the Clifford products

$$
\Theta^K = i^{\sigma_r}\,\Gamma^{A_1}\cdots\Gamma^{A_r},\quad \sigma_r=\tfrac{r(r-1)}{2}\bmod 2,\qquad \eta_K=\eta_{A_1A_1}\cdots\eta_{A_rA_r},\qquad \Theta^{()}=1_{16},
$$

so that every bilinear $\bar\psi\Theta^K\psi$ is real. Expanded about flat space, $g_{\mu\nu}=\eta_{\mu\nu}+h_{\mu\nu}$, the gravitational couplings of every field are the vertex $-\frac12 h_{\mu\nu}T^{\mu\nu}+O(h^2)$ with its energy–momentum tensor $T^{\mu\nu}$ (section 4). The Standard Model Lagrangian on the brane (frame of h, covariant derivative $D_\mu=\nabla_\mu-ig_sG^a_\mu T^a-igW^I_\mu\tau^I/2-ig'YB_\mu$) is

$$
\begin{aligned}
\mathcal{L}_{\rm SM} &= -\tfrac14 G^a_{\mu\nu}G^{a\mu\nu}-\tfrac14 W^I_{\mu\nu}W^{I\mu\nu}-\tfrac14 B_{\mu\nu}B^{\mu\nu}+(D_\mu\Phi)^\dagger(D^\mu\Phi)-\mu^2\Phi^\dagger\Phi-\lambda(\Phi^\dagger\Phi)^2\\
&\quad+\sum_f\bar f\,i\gamma^\mu D_\mu f-\big[\bar Q_L Y_u\tilde\Phi u_R+\bar Q_L Y_d\Phi d_R+\bar L_L Y_e\Phi e_R+\bar L_L Y_\nu\tilde\Phi\nu_R+\text{h.c.}\big],\\
G^a_{\mu\nu} &= \partial_\mu G^a_\nu-\partial_\nu G^a_\mu+g_s f^{abc}G^b_\mu G^c_\nu,\qquad W^I_{\mu\nu}=\partial_\mu W^I_\nu-\partial_\nu W^I_\mu+g\,\epsilon^{IJK}W^J_\mu W^K_\nu,\qquad B_{\mu\nu}=\partial_\mu B_\nu-\partial_\nu B_\mu,
\end{aligned}
$$

which contains the gauge–fermion vertices $g_s\bar q\gamma^\mu T^a qG^a_\mu$, $g\bar f_L\gamma^\mu(\tau^I/2)f_LW^I_\mu$, $g'Y\bar f\gamma^\mu fB_\mu$, the cubic and quartic gauge self-couplings in $-\frac14F^2$, the Higgs–gauge couplings in $|D\Phi|^2$, the Higgs self-couplings $-\mu^2|\Phi|^2-\lambda|\Phi|^4$

and the Yukawa couplings ($\tilde{\Phi} = i\sigma_2\Phi^*$). In the portal term L_i are the lepton doublets, ψ_s the brane-right-handed parts of dirac16complex at the brane, f any Standard Model fermion. Finally, the stabilizing potential used in sections 6–8 is

$$U(b) = U_0 + \frac{K_U}{2}\Big(\ln\frac{b}{b_0}\Big)^2, \qquad b^8 = \det(-g_{kl}) \quad (k,l = 4,\dots,7),$$

whose minimum b_0 fixes the size of the auxiliary space. One more interaction follows from these terms rather than being added: the RADION–BRANE coupling. The brane fields see $h = (b_0/b)^4 g_E$ with the four-dimensional Einstein-frame metric g_E, so that with the canonical radion $\phi = \sqrt{12}\,M_{\rm Pl}\ln(b/b_0)$

$$\mathcal{L}_{\phi W} = \frac{\phi}{\sqrt{3}\,M_{\rm Pl}}\,T^{\mu}{}_{\mu}\big|_{\rm brane}, \qquad T^{\mu}{}_{\mu} \supset m_f \bar{f} f,$$
$$T^{\mu}{}_{\mu} \supset -\frac{b_{\rm QCD}\,\alpha_s}{8\pi} G^a_{\mu\nu} G^{a\mu\nu} \quad (\text{trace anomaly},\ b_{\rm QCD} = 11 - \tfrac{2}{3}n_f).$$

4. The coupled field equations and the energy–momentum tensors (section 20.2)

Varying the action with respect to $g^{\mu\nu}$, $\bar{\psi}$ and the Standard Model fields gives the coupled field equations:

$$\sum_{k=0}^{3}\alpha_k E^{(k)\mu}{}_{\nu} = \kappa\Big[T_\psi + T_U + \frac{\delta^4(y)}{b^4}\big(T_f + T_{\rm SM} + T_P\big)\Big]^{\mu}{}_{\nu},$$
$$\big(i\Gamma^\mu D_\mu - m\big)\psi - \sum_{r=0}^{8} g_r \sum_{|K|=r}\big(\bar{\psi}\Theta_K\psi\big)\Theta^K\psi$$
$$= \frac{\delta^4(y)}{b^4}\Big[\sum_{i,s} y^*_{is}\,(\tilde{\Phi}^\dagger L_i)\,\chi_s + G^f_P\,(\bar{f}\gamma_\mu f)\,\Gamma^\mu\psi + \lambda_P\,\Phi^\dagger\Phi\,\psi\Big],$$
$$\nabla_\mu F^{a\mu\nu} = j^{a\nu}_{\rm SM} + j^{a\nu}_P,$$
$$0 = D_\mu D^\mu\Phi + \mu^2\Phi + 2\lambda(\Phi^\dagger\Phi)\Phi + (\text{Yukawa}) + \lambda_P(\bar{\psi}\psi)|_W\Phi,$$
$$i\gamma^\mu D_\mu f = (\text{Yukawa masses}) + G^f_P(\bar{\psi}\Gamma_\mu\psi)|_W\,\gamma^\mu f,$$

with $\Theta_K = \eta_K\Theta^K$, χ_s the portal spinor of the brane-right-handed component s, and j_P the portal currents. Every energy–momentum tensor is $T_{\mu\nu} = (2/\sqrt{|g|})\,\delta S/\delta g^{\mu\nu}$, i.e. $\delta S_{\rm matter} = \frac{1}{2}\int\sqrt{|g|}\,T_{\mu\nu}\delta g^{\mu\nu}$, with $T^0{}_0 = \varepsilon$. For the homogeneous metric every source is $T^{\mu}{}_{\nu} = \mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, -p_b, -p_b, -p_b)$. For matter whose comoving energy depends only on the scale factors at fixed occupation numbers, $W(a,b) = a^3b^4\varepsilon$, the action is $-\int nW\,dx_0$ and

$$\varepsilon = \frac{W}{a^3b^4}, \qquad p_a = -\frac{\partial_a W}{3a^2b^4}, \qquad p_b = -\frac{\partial_b W}{4a^3b^3}, \qquad \varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b) = 0.$$

The sources of the model in this form (all checked in section 20.2 of 7+1.nb):

source	$W(a,b)$	ε	p_a	p_b
stabilizing potential	$a^3b^4U(b)$	U	$-U$	$-U - \frac{b}{4}U'(b)$
brane tension (smeared, $\delta^4(y)\to 1/L_c^4$)	$a^3 f_W/L_c^4$	$f_W/V_\perp$	$-f_W/V_\perp$	0
brane fluid, $\rho_4 \propto a^{-3(1+w)}$	$a^3\rho_4/L_c^4$	$\rho_4/V_\perp$	$w\,\varepsilon$	0
one massless dirac16complex particle	$\omega = \sqrt{k_a^2/a^2 + k_b^2/b^2}$	$\omega/(a^3b^4)$	$k_a^2/(3a^2\omega\,a^3b^4)$	$k_b^2/(4b^2\omega\,a^3b^4)$

The minisuperspace action and the identities that its variation uses (EL_q is the Euler–Lagrange expression for the variable q, evaluated at $n = 1$) are

$$S = -\frac{1}{2\kappa}\int n\,a^3b^4\sum_k \alpha_k L_k\,dx_0 - \int nW\,dx_0,$$

$$\mathrm{EL}_n(na^3b^4L_k) = -2a^3b^4E^{(k)0}{}_0, \qquad \mathrm{EL}_a(na^3b^4L_k) = -6a^2b^4E^{(k)1}{}_1,$$
$$\mathrm{EL}_b(na^3b^4L_k) = -8a^3b^3E^{(k)4}{}_4,$$

and the variation gives exactly

$$\sum_k \alpha_k E^{(k)0}{}_0 = \kappa\varepsilon, \qquad \sum_k \alpha_k E^{(k)1}{}_1 = -\kappa p_a, \qquad \sum_k \alpha_k E^{(k)4}{}_4 = -\kappa p_b.$$

The dirac16complex gas used in the primordial regime is a sum of such particles: a collisionless gas with an isotropic comoving momentum distribution (the massless limit of the Kohn–Sham matter, valid while $m \ll H_a$ and the self-couplings are negligible); with $k_a = k\cos\theta$, $k_b = k\sin\theta$ and the weight $\cos^2\theta\sin^3\theta$ of the seven-dimensional momentum sphere,

$$\kappa\varepsilon = \frac{C_g}{a^3b^4}\langle\omega\rangle, \qquad \kappa p_a = \frac{C_g}{a^3b^4}\Big\langle\frac{\cos^2\theta}{3a^2\omega}\Big\rangle, \qquad \kappa p_b = \frac{C_g}{a^3b^4}\Big\langle\frac{\sin^2\theta}{4b^2\omega}\Big\rangle, \qquad \omega = \sqrt{\frac{\cos^2\theta}{a^2} + \frac{\sin^2\theta}{b^2}},$$
$$\langle f\rangle = \frac{\int_0^{\pi/2} f(\theta)\cos^2\theta\sin^3\theta\,d\theta}{\int_0^{\pi/2}\cos^2\theta\sin^3\theta\,d\theta},$$

evaluated with 128-point Gauss–Legendre quadrature in θ. The normalization $C_g = 3\times10^{-6}$ (units $\ell_2 = 1$, section 6) makes $\kappa\varepsilon = 10^{-6}\times 3H_a^2$ where $a = b = 1$ and $H_a = 1$.

5. The homogeneous equations, the radion and the four-dimensional constants (section 20.3)

With the closed forms of the Lovelock tensors of the cosmological metric ($H_a' = dH_a/dx_0$, $H_b' = dH_b/dx_0$):

$$\begin{aligned}
E^{(1)0}{}_0 &= 3\left(H_a^2 + 4H_aH_b + 2H_b^2\right),\\
E^{(1)1}{}_1 &= 2H_a' + 4H_b' + 3H_a^2 + 8H_aH_b + 10H_b^2,\\
E^{(1)4}{}_4 &= 3H_a' + 3H_b' + 6H_a^2 + 9H_aH_b + 6H_b^2,
\end{aligned}$$

$$\begin{aligned}
E^{(2)0}{}_0 &= -12H_b\left(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\right),\\
E^{(2)1}{}_1 &= -32H_a'H_aH_b - 48H_a'H_b^2 - 16H_b'H_a^2 - 96H_b'H_aH_b\\
&\quad - 48H_b'H_b^2 - 32H_a^3H_b - 136H_a^2H_b^2 - 192H_aH_b^3\\
&\quad - 60H_b^4,\\
E^{(2)4}{}_4 &= -12H_a'H_a^2 - 72H_a'H_aH_b - 36H_a'H_b^2 - 36H_b'H_a^2\\
&\quad - 72H_b'H_aH_b - 12H_b'H_b^2 - 12H_a^4 - 108H_a^3H_b\\
&\quad - 180H_a^2H_b^2 - 108H_aH_b^3 - 12H_b^4,
\end{aligned}$$

$$\begin{aligned}
E^{(3)0}{}_0 &= 360H_a^2H_b^3\left(4H_a + 3H_b\right),\\
E^{(3)1}{}_1 &= 72H_b^2\left(8H_a'H_aH_b + 2H_a'H_b^2 + 12H_b'H_a^2 + 8H_b'H_aH_b + 8H_a^3H_b + 19H_a^2H_b^2 + 8H_aH_b^3\right),\\
E^{(3)4}{}_4 &= 72H_aH_b\left(9H_a'H_aH_b + 6H_a'H_b^2 + 6H_b'H_a^2 + 9H_b'H_aH_b + 9H_a^3H_b + 17H_a^2H_b^2 + 9H_aH_b^3\right).
\end{aligned}$$

the three equations of section 4 are ordinary differential equations; the 11 and 44 equations are linear in (H_a', H_b') and the 00 equation is a constraint, preserved by the evolution. Written out, the constraint is

$$3(H_a^2 + 4H_aH_b + 2H_b^2) - 12\alpha_2H_b(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3) + 360\alpha_3H_a^2H_b^3(4H_a + 3H_b) - \Lambda = \kappa\varepsilon.$$

The radion is $\beta = \ln(b/b_0)$, $H_b = \beta'$. In Einstein gravity ($\alpha_2 = \alpha_3 = 0$) the 11 and 44 equations, with the constraint used to remove H_a^2, give exactly

$$\beta'' + 3H_a\beta' + 4\beta'^2 = \kappa\Big(\frac{\varepsilon}{6} - \frac{p_a}{2} + \frac{p_b}{3}\Big) + \frac{\Lambda}{3}.$$

With $U = U_0 + \frac{K_U}{2}\beta^2$ ($bU' = K_U\beta$) the potential enters the right-hand side through $\kappa(U/3 - bU'/12)$, i.e. as the restoring term $-(\kappa K_U/12)\beta$:

$$m_r^2 = \frac{\kappa K_U}{12}.$$

Static auxiliary space ($H_b = H_b' = 0$): the cubic term drops out ($E^{(3)} = 0$), the 00 and 11 equations are the Friedmann equations of the brane and the 44 equation fixes where the radion sits:

$$3H_a^2 = 8\pi G_4\,\rho_4 + \Lambda_4, \qquad H_a' = -\frac{\kappa}{2}(\varepsilon + p_a), \qquad 8\pi G_4 = \frac{\kappa}{V_\perp}, \qquad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp},$$
$$\frac{\kappa}{4}\,b\,U'(b) = \Lambda + \kappa U + \frac{2\kappa f_W}{V_\perp} + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2),$$

where ε, p_a, p_b are now the matter sources besides U and the tension. With U as above the equilibrium is $\beta_{\rm eq} = [\dots]/(3m_r^2)$, the bracket being the right-hand side of the last equation. **Einstein frame.** With $H_E = H_a + 2H_b$ (the expansion rate of $a_E = ab^2$) the constraint of Einstein gravity is exactly

$$3H_E^2 = 6\beta'^2 + \kappa\varepsilon \qquad \Longleftrightarrow \qquad 3M_{\rm Pl}^2 H_E^2 = \tfrac{1}{2}\phi'^2 + \rho, \qquad \phi = \sqrt{12}\,M_{\rm Pl}\,\beta,$$

so the canonical radion is $\phi = \sqrt{12}M_{\rm Pl}\beta$ and its coupling to the brane is the $\mathcal{L}_{\phi W}$ of section 3.

6. Regime I: the primordial Lovelock regime (section 21)

Units: $\ell_2 = \sqrt{\alpha_2} = 1$; then the vacuum equations contain the single parameter $\xi = \alpha_3/\alpha_2^2$, here $\xi = 10^{-6}$ (this separates the three Lovelock sub-regimes clearly; the crossover times below are given for any ξ). Λ is negligible ($\Lambda\ell_2^2 \approx (H_0/M_8)^2 \approx 10^{-92}$, section 9) and set to 0. If $\ell_2 = 1/M_8$ with $M_8 = 10$ TeV, the unit of time is 6.6×10^{-29} s.

6.1 The exact vacuum solution: an algebraic curve and one quadrature

With $u = H_b/H_a$ and $X = H_a^2$ the vacuum constraint divided by H_a^2 is

$$0 = P_E(u) + X\,P_G(u) + \xi X^2 P_C(u),$$
$$P_E = 3(2u^2 + 4u + 1), \quad P_G = -12u(u^3 + 12u^2 + 18u + 4), \quad P_C = 360u^3(3u + 4).$$

On $u \in (u_1, 0)$, $u_1 = -1 + 1/\sqrt{2}$, one has $P_E > 0$ and $P_C < 0$, so the quadratic has exactly one positive root and the vacuum solution is the explicit curve

$$H_a(u) = \sqrt{X(u)}, \qquad H_b(u) = u\sqrt{X(u)}, \qquad X(u) = \frac{P_G + \sqrt{P_G^2 - 4\xi P_C P_E}}{2\xi\,|P_C|},$$

along which the solution moves (H_a' from the 11 and 44 equations); the time is one quadrature,

$$\tau(H_a) = \int_{H_a}^{\infty} \frac{dH}{F(H)}, \qquad F = -H_a' > 0, \qquad p_{\rm eff} = -\frac{H_a^2}{H_a'} \;\; (= p \text{ for } a \propto \tau^p).$$

The two Kasner solutions ($a \propto \tau^p$, $b \propto \tau^q$) at the ends of the Lovelock hierarchy are

$$\text{Einstein:} \quad P_E(u_1) = 0, \quad u_1 = -1 + \tfrac{1}{\sqrt{2}}, \quad p_1 = \frac{1}{3 + 4u_1} = \frac{1 + 2\sqrt{2}}{7},$$
$$q_1 = u_1 p_1 = \frac{2 - 3\sqrt{2}}{14} \quad (3p_1 + 4q_1 = 1),$$
$$\text{Gauss–Bonnet:} \quad u_2^3 + 12u_2^2 + 18u_2 + 4 = 0,\; u_2 \in (-\tfrac{1}{3}, 0), \quad p_2 = \frac{3}{3 + 4u_2},$$
$$q_2 = u_2 p_2 \quad (3p_2 + 4q_2 = 3).$$

Numerically $u_1 = -0.2928932188$, $u_2 = -0.2695841903$, $p_1 = 0.5469181607$, $q_1 = -0.1601886205$, $p_2 = 1.561147624$, $q_2 = -0.4208607183$. The effective exponent along the curve is 0.54691816 at $u \to u_1$, 1.5537362 at $u = u_2$ and 1.0000018 at $u \to 0$ ($\xi = 10^{-6}$): it runs from the Einstein value through about the Gauss–Bonnet value to 1.

6.2 The accurate approximate solutions of the three Lovelock sub-regimes

Cubic regime ($\tau \ll \tau_{32}$). The series $H_a = 1/\tau + A_0 + A_1\tau + \dots$, $H_b = h_0 + h_1\tau + h_2\tau^2 + \dots$ solves the three equations order by order with (general α_2, α_3)

$$h_0 = -\sqrt{\frac{\alpha_2}{30\alpha_3}}, \qquad h_1 = \frac{\alpha_2}{16\alpha_3} - \frac{1}{32\alpha_2}, \qquad A_0 = \frac{\sqrt{30}\,(58\alpha_2^2 - 45\alpha_3)}{1920\,\alpha_2^{3/2}\alpha_3^{1/2}},$$
$$A_1 = -\frac{3647\,\alpha_2}{92160\,\alpha_3} - \frac{223}{6144\,\alpha_2} + \frac{645\,\alpha_3}{8192\,\alpha_2^3}, \qquad h_2 = \frac{\sqrt{30}\,(-1212\alpha_2^4 + 1180\alpha_2^2\alpha_3 + 225\alpha_3^2)}{307200\,\alpha_2^{5/2}\alpha_3^{3/2}},$$

i.e. for $\xi = 10^{-6}$: $h_0 = -182.5741858$, $h_1 = 62499.96875$, $h_2 = -2.160934549 \times 10^7$, $A_0 = 165.4577275$, $A_1 = -39572.51893$. Ordinary space expands like $a \propto \tau$ while b stays finite and shrinks at the finite rate $|h_0|$; the cubic and Gauss–Bonnet terms balance ($\alpha_3 E^{(3)} \approx -\alpha_2 E^{(2)}$).

Gauss–Bonnet regime ($\tau_{32} \ll \tau \ll \ell_2$). Expanding the curve about u_2: $p_{\rm eff} = p_2 + f_1/H_a^2 + f_2\,\xi H_a^2 + \dots$ with $f_1 = -0.6988734676$, $f_2 = -0.7853407142$ (units $\alpha_2 = 1$), so that

$$\tau = \frac{p_2}{H_a} + \frac{f_1}{3H_a^3} - f_2\,\xi H_a + s_2, \qquad a \propto (\tau - s_2)^{p_2}, \qquad b \propto (\tau - s_2)^{q_2}, \qquad H_{a,\rm mid} = \Big(\frac{f_1}{f_2\,\xi}\Big)^{1/4},$$

with the time shift $s_2 = -0.00098646164$ fixed at $H_{a,\rm mid} = 30.713910$.

Einstein regime ($\tau \gg \ell_2$). Expanding about u_1: $u = u_1 + d_1 H_a^2 + \dots$, $d_1 = \alpha_2(16 - 11\sqrt{2})/4$ (0.1109127035), and $p_{\rm eff} = p_1 + e_1 H_a^2 + \dots$, $e_1 = \alpha_2(1 + 9\sqrt{2})/7$ (1.961131723), so that

$$\tau = \frac{p_1}{H_a} - e_1 H_a + s_1, \qquad H_a = \frac{p_1}{\tau - s_1} - \frac{e_1 p_1^2}{(\tau - s_1)^3} + \dots, \qquad a \propto (\tau - s_1)^{p_1}, \qquad b \propto (\tau - s_1)^{q_1},$$

with $s_1 = \tau(u_2) - p_1/H_a(u_2) + \int_0^{H_a(u_2)} \big(1/F - p_1/H^2\big)dH = 2.079588493$ from the exact quadrature ($\tau(u_2) = 0.1593549798$, $H_a(u_2) = 9.721560116$).

Crossovers and accuracy. The cubic → Gauss–Bonnet crossover is at $\tau_{32} \approx |q_2|/|h_0| = |q_2|\sqrt{30\xi}\,\ell_2$, the Gauss–Bonnet → Einstein crossover at $\tau_{21} \approx \ell_2$; on the exact solution $\tau(H_a = 10^3) = 0.00114712$ and $\tau(H_a = 1) = 1.39471$. Against the exact solution (integrated from cubic-series data at $\tau = 10^{-8}$ in the variable $\sigma = -\ln H_a$, and checked against the quadrature), the largest relative errors of $\tau(H_a)$ are:

approximation	**window**	**largest relative error**
cubic series (A_0, A_1; h_0, h_1, h_2)	$H_a \geq 10^4$	6.7×10^{-6}
Gauss–Bonnet Kasner with corrections and s_2	$3 \leq H_a \leq 300$	0.00085
Einstein Kasner with correction and s_1	$H_a \leq 0.1$	0.00032
best of the three, in the crossover windows	$10^{-3.5} \leq H_a \leq 10^8$	0.10

Inside each regime the formulas are accurate to better than 10^{-3}; in the two crossover windows the exact curve-plus-quadrature solution of section 6.1 is used.

6.3 Coupled to dirac16complex: the matter-dominated cubic regime

Couple the gas of section 4 (isotropic, $a = b = 1$, at $H_a = 1$, where $\kappa\varepsilon = 10^{-6} \times 3H_a^2$). Near $\tau = 0$ any fluid with $p_a = w_a\varepsilon$ ($w_a < 1$) wins over the vacuum terms. With b finite and $H_a = p/\tau$ the leading orders of the three equations are

$$\tau^3\big(\textstyle\sum \alpha_k E^{(k)0}{}_0\big) \to 48\,h\,p^3(30\alpha_3 h^2 - \alpha_2), \qquad \tau^3\big(\textstyle\sum \alpha_k E^{(k)1}{}_1\big) \to 32\,h\,p^2(p-1)(18\alpha_3 h^2 - \alpha_2),$$
$$\tau^4\big(\textstyle\sum \alpha_k E^{(k)4}{}_4\big) \to p^3(p-1)(648\alpha_3 h^2 - 12\alpha_2),$$

so that (p_b is of lower order)

$$h_* = -\sqrt{\frac{\alpha_2}{54\alpha_3}}, \qquad p = \frac{1}{1 + w_a}, \qquad \kappa\varepsilon\,\tau^3 \to \frac{64}{3}\,p^3\alpha_2|h_*|, \qquad \alpha_2 E^{(2)} : \alpha_3 E^{(3)} : \kappa\varepsilon \to \frac{9}{4} : -\frac{5}{4} : 1.$$

For the gas $w_a \to 1/3$ as $a/b \to 0$: $a \propto \tau^{3/4}$ and $\kappa\varepsilon\tau^3 \to 9\alpha_2|h_*|$. The coupled numerical solution gives at $H_a = 10^{15}$: $p_{\rm eff} = 0.75000033$, $H_b = -136.08292$ ($h_* = -136.08276$), $\kappa\varepsilon\tau^3 = 1224.7435$ ($9|h_*| = 1224.7449$), $\alpha_2 E^{(2)}/\kappa\varepsilon = 2.250007$, $\alpha_3 E^{(3)}/\kappa\varepsilon = -1.250007$.

The size of each term. In a Kasner regime the monomials of the dominant curvature term cancel each other (that is what makes $P_E(u_1)=0$ or $P_G(u_2)=0$), so its net value would hide it. The SIZE of a curvature term of the 00 equation is therefore the sum of the absolute values of its monomials in H_a and H_b, and the GAS FRACTION is $\kappa\varepsilon$ divided by the sum of all sizes:

$$S_1 = 3\big(H_a^2 + 4H_a|H_b| + 2H_b^2\big), \qquad S_2 = 12\alpha_2|H_b|\big(4H_a^3 + 18H_a^2|H_b| + 12H_aH_b^2 + |H_b|^3\big),$$
$$S_3 = 360\alpha_3 H_a^2|H_b|^3\big(4H_a + 3|H_b|\big), \qquad f_{\rm gas} = \frac{\kappa\varepsilon}{S_1+S_2+S_3+\kappa\varepsilon} \quad (H_a>0).$$

Moving forward in time the gas fraction DECREASES in the cubic regime ($\varepsilon\propto\tau^{-4}$ against τ^{-3}) and in the Gauss–Bonnet regime ($\varepsilon\propto\tau^{-4(p_2+q_2)}$ against τ^{-4}), so the vacuum regimes of section 6.2 follow: $(p_{\rm eff}, f_{\rm gas}) = (0.999318,\, 0.00119)$ at $H_a = 3\times10^5$, $(1.55967,\, 1.85\times10^{-7})$ at $H_a=30$, $(0.547086,\, 2.83\times10^{-5})$ at $H_a=10^{-2}$. In the Einstein regime the auxiliary momenta are blue-shifted ($p_b\to\varepsilon/4$, $p_a\to0$) and $\kappa\varepsilon/(3H_a^2)$ — and with it $f_{\rm gas}$, since there $S_1\approx 3(1+4|u_1|+2u_1^2)H_a^2$ — GROWS like $\tau^{2-3p_1-5q_1}=\tau^{1.16019}$ (measured between $H_a=10^{-2}$ and 10^{-4}: 1.16983).

6.4 Without a stabilizing potential the auxiliary space re-expands

Einstein gravity with a fluid $p_a = w_a\varepsilon$, $p_b = w_b\varepsilon$ has the power-law solutions $a\propto\tau^P$, $b\propto\tau^Q$ with $3P(1+w_a)+4Q(1+w_b)=2$ and

$$\tau^2 E^{(1)0}{}_0 = 3(P^2+4PQ+2Q^2) = \mathcal{C}, \qquad \tau^2 E^{(1)1}{}_1 = 3P^2+8PQ-2P+10Q^2-4Q = -w_a\mathcal{C},$$
$$\tau^2 E^{(1)4}{}_4 = 3(2P^2+3PQ-P+2Q^2-Q) = -w_b\mathcal{C} \qquad (\mathcal{C}=\kappa\varepsilon\tau^2):$$

fluid (w_a, w_b)	(P, Q)	$\mathcal{C}=\kappa\varepsilon\tau^2$	**meaning**
$(0, \frac14)$ (auxiliary radiation)	$(0, \frac25)$	$\frac{24}{25}$	ordinary space stops, the auxiliary space expands
$(\frac13, 0)$ (brane radiation)	$(\frac12, 0)$	$\frac34$	ordinary radiation era, b frozen
$(\frac17, \frac17)$ (isotropic radiation)	$(\frac14, \frac14)$	$\frac{21}{16}$	isotropic expansion

The coupled solution confirms the first line: at $H_a=10^{-14}$, $\tau H_a = 0.0003149$, $\tau H_b = 0.399995$, $p_a/\varepsilon = 1.87\times10^{-6}$, $p_b/\varepsilon = 0.249999$. Once the gas dominates, the 7+1 universe DECOMPACTIFIES unless a stabilizing potential holds b (section 7.1). The Standard Model plasma cannot do it: confined to the brane at $T\le M_8$, its smeared density is a fraction $(\pi^2 g_*/30)M_8^4/(3M_{\rm Pl}^2M_8^2) = 1.97\times10^{-28}$ of the curvature terms at $H=M_8$ ($M_{\rm Pl}=2.435\times10^{18}$ GeV, $g_*=106.75$).

6.5 Pair creation of dirac16complex in the primordial regime

A massless mode on the Einstein–Kasner background obeys $i\,dc/d\tau = (K\tau^{-p_1}\sigma_1 + K\tau^{-q_1}\sigma_2)c$ in the scaling variable K (Pauli matrices $\sigma_{1,2}$); the Bogoliubov coefficient $F(K)=|\beta|^2$ is the probability that a positive-energy state at early times ends in a negative-energy state at late times. $F(1)=0.010749346$, $F(8)=7.10407\times10^{-6}$, $F(16)=8.15730\times10^{-7}$: for large K it falls as a POWER,

$$F(K)\propto K^{-2(p_1-q_1)/(1-p_1)} = K^{-(3\sqrt2+2)/2} = K^{-3.1213\ldots} \qquad \text{(measured between } K=8 \text{ and } 16:\ -3.12248),$$

because near $x_0=0$ the phase grows only like $K\tau^{1-p_1}$ while the direction of the mode Hamiltonian turns like $\tau^{p_1-q_1}$: the non-adiabaticity comes from the singularity itself. The modes that become non-adiabatic at the time τ have physical momenta of order $1/\tau$, so their physical number density is of order τ^{-7} (seven space dimensions); per comoving volume, with $a^3b^4=\tau^{3p_1+4q_1}=\tau$, the number of pairs created per logarithmic interval of time grows like

$$\frac{dN_{\rm comoving}}{d\ln\tau}\propto\tau^{-7}\,\tau^{3p_1+4q_1}=\tau^{-6} \qquad (\tau\to0):$$

the production is dominated by the earliest, Planckian times, where it gives a gas fraction of order one — the gas of section 6.3.

7. Regime II: the modern regime (section 22)

7.1 The stabilizing potential catches the auxiliary space

Add $U = \frac{K_U}{2}\beta^2$ ($U_0 = 0$, $\kappa K_U = 12m_r^2$, $m_r = 10^{-3}/\ell_2$) to the coupled solution of section 6.3 at $H_a = 10^{-2}$ ($\tau = 56.7505$, $\beta = 0.459596$), with b_0 where the Kasner contraction would arrive at $\tau = 1/m_r$. When the expansion rate falls below m_r the radion is caught and oscillates; its energy then behaves as four-dimensional matter, and the radion equation of section 5 in a matter-dominated universe, $\beta'' + \frac{2}{\tau}\beta' + m_r^2\beta = 0$, has the exact solution that is the accurate approximation here:

$$\beta \approx \frac{A\sin(m_r\tau + \vartheta)}{m_r(\tau - \tau_E)}, \qquad H_E = H_a + 2H_b \approx \frac{2}{3(\tau - \tau_E)}, \qquad \text{corrections } O\big(1/(m_r\tau)\big).$$

With τ_E from the average of H_E over the last period ($\tau_E = -1564.358$) and A, ϑ by linear least squares on $10^5 \le \tau \le 3\times 10^5$: $A = 0.4693305$, $\vartheta = 2.136009$, largest deviation 0.013 of the amplitude; $\langle H_E\rangle(\tau - \tau_E) = 0.666942$ and 0.666736 at $\tau = 10^5$ and 2×10^5 (exact value 2/3). The gas share stays constant (0.008596, 0.008728): with frozen auxiliary momenta (Kaluza–Klein masses) the gas is matter too. The radion carries 0.9913 of the energy: the cosmological MODULI PROBLEM of the minimal model.

7.2 Radion decay and the Kaluza–Klein channels

In physical units ($M_8 = 10$ TeV, $M_{\rm Pl} = 2.435\times 10^{18}$ GeV, $V_\perp = M_{\rm Pl}^2/M_8^6 = 5.92923\times 10^{12}$ GeV^{-4}, $\ell_\perp = V_\perp^{1/4} = 3.07919\times 10^{-13}$ m, lowest Kaluza–Klein mass $2\pi/\ell_\perp = 0.00402652$ GeV) the coupling $\mathcal{L}_{\phi W}$ of section 3 gives

$$\Gamma(\phi \to f\bar f) = \frac{N_c\, m_f^2\, m_r}{24\pi M_{\rm Pl}^2}\Big(1 - \frac{4m_f^2}{m_r^2}\Big)^{3/2}, \qquad \Gamma(\phi \to gg) = \frac{b_{\rm QCD}^2\alpha_s^2\, m_r^3}{96\pi^3 M_{\rm Pl}^2}, \qquad T_{\rm RH} = \Big[\frac{\Gamma M_{\rm Pl}}{(\pi^2 g_*/90)^{1/2}}\Big]^{1/2}.$$

For $m_r = 10$ GeV (b, c, τ, μ pairs and gluons, $\alpha_s = 0.18$, $n_f = 5$, $g_* = 10.75$) the brane channels give $\Gamma_{\rm brane} = 4.572\times 10^{-37}$ GeV, a lifetime of 1.440×10^{12} s, and would reheat the brane only to $T_{\rm RH} = 1.013\times 10^{-9}$ GeV — far below the 4 MeV that nucleosynthesis needs.

Decay into the bulk. The radion also couples to every Kaluza–Klein mode of the bulk fields. A mode of dirac16complex with Kaluza–Klein momentum $\mu_n = 2\pi|n|/\ell_\perp$ ($n \in \mathbb{Z}^4$) has, in the four-dimensional Einstein frame ($h = e^{-4\beta}g_E$), the mass

$$M_E = e^{-2\beta}\sqrt{m^2 + \mu_n^2 e^{-2\beta}}, \qquad g_n = -\frac{\partial M_E/\partial\beta}{\sqrt{12}M_{\rm Pl}}\Big|_{\beta=0} = \frac{2M_n + \mu_n^2/M_n}{\sqrt{12}\,M_{\rm Pl}}, \qquad M_n = \sqrt{m^2 + \mu_n^2},$$

(a brane fermion has the coupling $m_f/(\sqrt{3}M_{\rm Pl})$ of $\mathcal{L}_{\phi W}$, the case $\mu_n = 0$). A scalar coupled as $-g\phi\bar\chi\chi$ decays at the rate $g^2 m_r(1 - 4M^2/m_r^2)^{3/2}/(8\pi)$; for $m \ll \mu_n$ each Kaluza–Klein level (four brane Dirac spinors) gives $\Gamma_n = 3\mu_n^2 m_r(1 - 4\mu_n^2/m_r^2)^{3/2}/(8\pi M_{\rm Pl}^2)$. The sum over the 1.173×10^{13} momenta with $2\mu_n < m_r$, done shell by shell with the number $r_4(k) = 8\sigma(k) - 32\sigma(k/4)$ of lattice vectors with $|n|^2 = k$ (σ the divisor sum; $\sigma(k/4) = 0$ unless $4 \mid k$; checked against a direct count for $k \le 60$), equals its continuum limit:

$$\Gamma_{\rm bulk} = \sum_n \Gamma_n \approx \frac{3m_r}{8\pi M_{\rm Pl}^2}\,\frac{V_\perp}{8\pi^2}\Big(\frac{m_r}{2}\Big)^6\frac{B(3,\frac52)}{2}, \qquad B(3,\tfrac52) = \frac{\Gamma(3)\Gamma(\frac52)}{\Gamma(\frac{11}{2})} = \frac{16}{315},$$

$\Gamma_{\rm bulk} = 5.99917\times 10^{-24}$ GeV (continuum 5.99917×10^{-24} GeV), a lifetime of 0.110 s: 1.3×10^{13} times faster than the brane channels (the Kaluza–Klein gravitons add channels of the same strength; for $m > m_r/2$ the dirac16complex channels close). The radion decays into the BULK: its energy goes to Kaluza–Klein modes, which return it to the brane only through couplings suppressed by the volume (a Kaluza–Klein graviton of mass M decays into brane fields at a rate of order $M^3/M_{\rm Pl}^2$). This is the cosmological moduli problem of the minimal model in the form it takes with large extra dimensions; the model needs the radion and bulk energy left by regime I to be removed before nucleosynthesis (for example by a later period of inflation on the brane, which is not part of the model). The modern solutions below start after such a reheating, below the normalcy temperature $T_* \approx (M_8^6/M_{\rm Pl})^{1/5} = 13.26$ GeV (above it the brane plasma would lose its energy to bulk gravitons faster than it cools by expansion; an order-of-magnitude estimate).

7.3 The Standard Model plasma: $g_*(T)$, $g_{*s}(T)$ and the time–temperature relation

With b static the brane obeys $3M_{\rm Pl}^2H_a^2=\rho_4+M_{\rm Pl}^2\Lambda_4$ and $H_a'=-(\rho_4+p_4)/(2M_{\rm Pl}^2)$. In the radiation era $\rho_4=(\pi^2/30)g_*T^4$ and the entropy $s=(2\pi^2/45)g_{*s}T^3$ is conserved, $sa^3=\text{const}$:

$$H_a=\Big(\frac{\pi^2 g_*}{90}\Big)^{1/2}\frac{T^2}{M_{\rm Pl}},\qquad \ln a=-\ln T-\tfrac{1}{3}\ln g_{*s}(T)+\text{const},$$
$$t(T)=t(T_1)+\int\frac{d\ln a}{H_a}=t(T_1)+\int_T^{T_1}\frac{d\tilde T}{\tilde T}\,\frac{1+\frac{1}{3}\,d\ln g_{*s}/d\ln\tilde T}{H_a(\tilde T)}.$$

Each species i ($x=m_i/T$, degeneracy g_i; bosons $+$, fermions with alternating signs) contributes

$$\rho_i=\frac{g_iT^4}{2\pi^2}\sum_{n\ge1}(\pm1)^{n+1}\frac{3(nx)^2K_2(nx)+(nx)^3K_1(nx)}{n^4},$$
$$p_i=\frac{g_iT^4}{2\pi^2}\sum_{n\ge1}(\pm1)^{n+1}\frac{(nx)^2K_2(nx)}{n^4},\qquad s_i=\frac{\rho_i+p_i}{T},$$

$g_*=30\rho/(\pi^2T^4)$, $g_{*s}=45s/(2\pi^2T^3)$, for all Standard Model particles with their masses; the QCD crossover is modelled by the switch $S(T)=\frac{1}{2}[1+\tanh((T-0.160\,\text{GeV})/0.030\,\text{GeV})]$ between quarks and gluons and the lightest hadrons ($\pi,K,\eta,\rho,\omega,K^*,\eta',p,n$) — this model differs from the lattice equation of state by up to about 10% near the crossover; the neutrinos decouple while $e^\pm$ are relativistic, $T_\nu=T\,(g_{\gamma e}(T)/\frac{11}{2})^{1/3}$, and are scaled by $N_{\rm eff}/3=3.044/3$. The integral is evaluated on 481 points equally spaced in $\ln T$ (10 GeV to 10 keV), starting from $t(T_1)=1/(2H_a(T_1))$ at $T_1=10$ GeV; on each step $1/H_a$ is taken to be an exponential function of $\ln a$ (exact where g_* and g_{*s} are constant), so a step adds $\Delta\ln a\,(1/H_{i+1}-1/H_i)/\ln(H_i/H_{i+1})$ (halving the grid changes $t(10\,\text{keV})$ by 3.6×10^{-7}).

T	**1 TeV**	**10 GeV**	**1 GeV**	**300 MeV**	**100 MeV**	**10 MeV**	**1 MeV**	**100 keV**	**1 keV**
g_*	106.794	86.2963	76.4179	64.6458	18.3152	10.8372	10.6336	4.32702	3.38300
g_{*s}	106.778	86.2045	75.5771	64.2172	17.9496	10.8352	10.6361	4.81438	3.93746

T	**1 GeV**	**100 MeV**	**10 MeV**	**1 MeV**	**100 keV**	**10 keV**
t (s)	2.79410×10^{-7}	6.12354×10^{-5}	0.00739300	0.743187	118.840	13161.7
$1/(2H_a)$ divided by t	0.990757	0.923416	0.994323	0.998548	0.978928	0.999641

The instantaneous approximation $t\approx1/(2H_a)$ is good to about 1% where g_* is nearly constant; where g_* changes quickly it is off by 7.7% at 100 MeV (QCD crossover) and by 2.1% at 100 keV ($e^\pm$ annihilation).

7.4 The late universe: an accurate analytic solution and the age

Below about 1 MeV the brane contains photons and neutrinos (Ω_r from $T_{\rm CMB}=2.7255$ K and $N_{\rm eff}=3.044$: $\Omega_r=\Omega_\gamma[1+\frac{7}{8}(\frac{4}{11})^{4/3}N_{\rm eff}]$, $\Omega_\gamma=(\pi^2/15)T_{\rm CMB}^4/(3H_0^2M_{\rm Pl}^2)$), matter ($\Omega_m=0.315$; its cold dark matter would be dirac16complex if dirac16complex is the dark matter) and Λ_4 ($\Omega_\Lambda=1-\Omega_m-\Omega_r$), with $H_0=67.4$ km/s/Mpc (Planck 2018). The Friedmann equation $H_a=H_0(\Omega_ra^{-4}+\Omega_ma^{-3}+\Omega_\Lambda)^{1/2}$ has exact two-component solutions, and their composite

$$t_{rm}(a)=\frac{2}{3H_0\Omega_m^2}\Big[(\Omega_ma-2\Omega_r)\sqrt{\Omega_r+\Omega_ma}+2\Omega_r^{3/2}\Big],\qquad t_{m\Lambda}(a)=\frac{2}{3H_0\Omega_\Lambda^{1/2}}\,\text{arcsinh}\Big(\sqrt{\tfrac{\Omega_\Lambda}{\Omega_m}}\,a^{3/2}\Big),$$
$$t(a)\approx t_{rm}(a)+t_{m\Lambda}(a)-t_m(a),\qquad t_m(a)=\frac{2}{3H_0\Omega_m^{1/2}}\,a^{3/2},$$

is accurate to 0.00022 at every time (the error is the product of the small radiation and Λ corrections). Numbers: $\Omega_r=9.210\times10^{-5}$, $\Omega_\Lambda=0.684908$, $z_{\rm eq}=\Omega_m/\Omega_r-1=3419.35$, age $=\int_0^1 da/(aH_a)=13.7907$ Gyr (Planck 2018: 13.787 ± 0.020).

The radion today sits at $\beta_{\rm eq} = [\Lambda_4 + \kappa f_W/V_\perp + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2)]/(3m_r^2)$: for $m_r = 10$ GeV, 1.74×10^{-86} without brane tension and 5.62×10^{-24} with $f_W = M_8^4$ — so $G_4 = G_8/V_\perp$ is constant. The Gauss–Bonnet and cubic corrections are $\alpha_2 H_0^2 = 2.07 \times 10^{-92}$ and $\alpha_3 H_0^4 = 4.27 \times 10^{-190}$ ($\alpha_2 = 1/M_8^2$, $\alpha_3 = 10^{-6}/M_8^4$).

8. Regime III: the far future (section 23)

Once matter has diluted, Λ_4 dominates: $H_\infty = H_0\Omega_\Lambda^{1/2} = 1.8077 \times 10^{-18}$ s^{-1} (1.1898×10^{-42} GeV), Gibbons–Hawking temperature $H_\infty/(2\pi) = 2.198 \times 10^{-30}$ K. For matter $+\Lambda$ the solution is exact,

$$a(t) = \Big(\frac{\Omega_m}{\Omega_\Lambda}\Big)^{1/3} \sinh^{2/3}\Big(\frac{3}{2}H_\infty t\Big), \qquad H_a(t) = H_\infty \coth\Big(\frac{3}{2}H_\infty t\Big) \to H_\infty + 2H_\infty e^{-3H_\infty t},$$

and agrees with the full solution (radiation included) to 10^{-4} already at $a = 3$. Pairs of dirac16complex are created with the Fermi–Dirac occupation $|\beta|^2 = 1/(e^{2\pi M/H} + 1)$; $2\pi M/H_\infty = 5.28 \times 10^{33}$ for $M = 1$ eV: none are.

The radion in de Sitter. Linearizing the full Einstein–Lovelock equations (with $U = \frac{K_U}{2}\beta^2$) about $H_a = H$, $H_b = 0$, $\beta = \beta_{\rm eq}$, with δH_a removed by the constraint, gives $\beta'' + \gamma\beta' + m_{\rm eff}^2(\beta - \beta_{\rm eq}) = 0$ with (for $\beta_{\rm eq} \to 0$)

$$\frac{\gamma}{3H} = \frac{1 - 16\alpha_2 H^2 + 48\alpha_2^2 H^4 - 48\alpha_3 H^4 + 512\alpha_2^3 H^6 + 192\alpha_2\alpha_3 H^6 - 1024\alpha_2^4 H^8}{(1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4)^2},$$
$$\frac{m_{\rm eff}^2}{m_r^2} = \frac{1}{1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4}, \qquad 3m_r^2\beta_{\rm eq} = 3H^2 - 12\alpha_2 H^4 + O(\beta_{\rm eq}^2).$$

For $\alpha_2 = \alpha_3 = 0$: $\gamma = 3H$, $m_{\rm eff} = m_r$. A numerical test in units $\ell_2 = 1$ ($\Lambda = 3 \times 10^{-4}$, $m_r = 0.1$, $\xi = 10^{-6}$, start 0.05 above equilibrium) relaxes to $\beta = 0.01020407809$, $H_a = 0.01010358670$, the solution of $3H^2 = \Lambda + 6m_r^2\beta^2$, $3m_r^2\beta = 3H^2 - 12H^4$ (0.01020407809, 0.01010358670). In the physical far future $\alpha_2 H_\infty^2 \approx 10^{-92}$ and $\beta_{\rm eq} \approx H_\infty^2/m_r^2$: the auxiliary space stays frozen. Without the stabilizing potential the far future would instead be eight-dimensional de Sitter, $a = b = e^{Hx_0}$, an exact vacuum solution when all three equations read

$$21H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 = \Lambda;$$

the auxiliary space would then inflate and $G_4 \propto b^{-4}$ would fall like e^{-4Hx_0}, which experiments exclude ($|\dot G/G|$ is less than a few percent of H_0).

9. The size of every interaction in the three regimes

Regime I. The shares of the four terms of the 00 equation at four points of the coupled primordial solution of section 6.3: each size (S_1, S_2, S_3 of section 6.3, and $\kappa\varepsilon$) divided by $S_1 + S_2 + S_3 + \kappa\varepsilon$:

point of regime I	**Einstein** S_1	**Gauss–Bonnet** S_2	**cubic** S_3	**gas** $\kappa\varepsilon$
cubic ($H_a = 3 \times 10^5$)	5.73×10^{-10}	0.501	0.498	0.00119
Gauss–Bonnet ($H_a = 30$)	0.000235	0.999	0.000964	1.85×10^{-7}
Einstein ($H_a = 10^{-2}$)	0.999	0.000516	6.27×10^{-14}	2.83×10^{-5}
near the singularity ($H_a = 10^{15}$)	2.30×10^{-19}	0.500	0.278	0.222

In the cubic regime the Gauss–Bonnet and cubic terms balance each other; in the Gauss–Bonnet and Einstein regimes one term carries the 00 equation alone (its monomials cancel among themselves, which is what fixes u_2 and u_1); near the singularity the gas joins the Gauss–Bonnet and cubic terms in the ratio $\alpha_2 E^{(2)} : \alpha_3 E^{(3)} : \kappa\varepsilon = 9/4 : -5/4 : 1$ of section 6.3.

All interactions. Every term of the action of section 3 with its weight in the field equations of each regime ($M_8 = 10$ TeV, $\alpha_2 = 1/M_8^2$, $\alpha_3 = 10^{-6}/M_8^4$, $m_r = 10$ GeV, $f_W = M_8^4$, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV; regime I at $H = 1/\ell_2 = M_8$; "estimate" marks an order-of-magnitude statement):

interaction	regime I (primordial)	regime II (today)	regime III (far future)
S_0: $\Lambda_4/(3H^2)$	$\Omega_\Lambda \alpha_2 H_0^2 = 1.4 \times 10^{-92}$ (if $\Lambda \approx \Lambda_4$)	$\Omega_\Lambda = 0.6849$	$\to 1$
S_1: Einstein term	dominant for $\tau \gg \ell_2$	dominant (four-dimensional Friedmann equations)	dominant
S_2: $\alpha_2 H^2$	dominant for $\tau_{32} \ll \tau \ll \ell_2$	2.07×10^{-92}	1.42×10^{-92}
S_3: $\alpha_3 H^4$	dominant for $\tau \ll \tau_{32}$	4.27×10^{-190}	2.00×10^{-190}
S_ψ: dirac16complex (gas; matter)	see the table above; dominant near $\tau = 0$	$\Omega_m = 0.315$ (its cold dark matter, $\Omega_c = \Omega_c h^2/h^2 = 0.120/0.674^2 = 0.264$, if it is the dark matter)	$\to 0$ like $e^{-3H_\infty t}$
S_m: m/H	$\ll 1$ (massless gas)	$\gg 1$ (cold matter)	$\gg 1$
S_{self}: $g_r n^{6/7} \approx (k_{\text{phys}}/M_8)^6$ (estimate)	< 1 below the eight-dimensional Planck scale	negligible	negligible
S_U: radion displacement β	catches b when $H \approx m_r$, then oscillates	$\beta_{\text{eq}} = 1.74 \times 10^{-86}$	$\beta_{\text{eq}} \approx H_\infty^2/m_r^2 = 1.42 \times 10^{-86}$
S_f: $\kappa f_W/(3H^2 V_\perp)$	5.62×10^{-30}	part of Λ_4; shifts β_{eq} by 5.62×10^{-24}	part of Λ_4
S_{SM}: brane plasma	1.97×10^{-28} (at $T = M_8$)	$\Omega_r = 9.21 \times 10^{-5}$; baryons $\Omega_b \approx 0.049$	$\to 0$ like $e^{-4H_\infty t}$
S_P: portals (estimate)	negligible (on the brane)	production of dirac16complex (freeze-in)	negligible
$\mathcal{L}_{\phi W}$: radion–brane coupling	radion not yet caught	brane channels alone: $T_{\text{RH}} = 1.01 \times 10^{-9}$ GeV ($m_r = 10$ GeV)	—
radion–bulk couplings g_n	radion not yet caught	decay into bulk Kaluza–Klein modes in 0.11 s	—
pair creation, $2\pi M/H$ ($M = 1$ eV)	of order one (Planckian times)	4.37×10^{33}	5.28×10^{33}

10. What the solutions mean

- **The primordial regime is under control.** Its vacuum solution is exact (section 6.1), and its three Lovelock sub-regimes have accurate closed forms. Near $x_0 = 0$ any dirac16complex gas makes the universe matter-dominated with $a \propto \tau^{3/4}$ and a finite contraction rate h_* of the auxiliary space; pair creation from the singularity supplies such a gas.
- **Dynamical compactification is temporary.** In the Gauss–Bonnet and Einstein regimes the auxiliary space contracts ($q_2, q_1 < 0$) while ordinary space expands; but once the bulk gas dominates, the auxiliary space re-expands ($b \propto \tau^{2/5}$). The Standard Model brane, at $T \leq M_8$, is too dilute in the bulk to stop it. A stabilizing potential is a necessary part of the model.
- **Stabilization costs a moduli problem.** The potential catches b when $H \approx m_r$, but the radion then carries almost all the energy (0.991). Through brane fields alone it would decay far too late for nucleosynthesis; with its 10^{13} channels of gravitational strength into the bulk Kaluza–Klein modes it decays within 0.11 s, but its energy then sits in the bulk and returns to the brane only through volume-suppressed couplings. The minimal model needs an additional dilution and reheating of the brane after stabilization.
- **The modern regime is standard cosmology on the brane.** With b fixed (to 10^{-86}–10^{-24}) gravity is four-dimensional with $G_4 = G_8/V_\perp$, the Lovelock corrections are 10^{-92} and smaller, and the brane Friedmann solution reproduces the age of the universe; dirac16complex enters only through its energy density (cold matter).
- **The far future is de Sitter** with a stable, frozen auxiliary space; without the stabilizer it would be eight-dimensional de Sitter (excluded). No pairs are created.
- **Validity.** The classical solutions need curvatures below the eight-dimensional Planck scale ($\tau \gg \kappa^{1/6}$), which requires $\ell_2 \gg \kappa^{1/6}$ for the cubic and Gauss–Bonnet regimes to be classical; the brane plasma needs $T < M_8$ and $T < T_*$; the massless-gas description of dirac16complex needs $m \ll H_a$ and negligible self-couplings.

11. Verification

Every statement above is a check of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_coupled.py (numpy, scipy, sympy, mpmath), which uses its own closed forms (from the independent Lovelock engine tools/lovelock71.py), its own quadrature, root finding and Runge–Kutta integrator (DOP853). The numbers printed by both programs:

quantity	**7+1.nb**	**Python**	**max. relative difference**	**tolerance**
Kasner constants $u_1, u_2, p_1, q_1, p_2, q_2$	-0.2928932, -0.2695842, 0.5469182, -0.1601886, 1.561148, -0.4208607	-0.2928932, -0.2695842, 0.5469182, -0.1601886, 1.561148, -0.4208607	0	1×10^{-14}
$p_{\rm eff}$ at $u \to u_1$, $u = u_2$, $u \to 0$	0.5469182, 1.553736, 1.000002	0.5469182, 1.553736, 1.000002	0	1×10^{-10}
cubic-regime coefficients h_0, h_1, h_2, A_0, A_1	-182.5742, 62499.97, -2.160935×10^7, 165.4577, -39572.52	-182.5742, 62499.97, -2.160935×10^7, 165.4577, -39572.52	0	1×10^{-13}
Gauss–Bonnet coefficients f_1, f_2	-0.6988735, -0.7853407	-0.6988735, -0.7853407	1.6×10^{-16}	1×10^{-9}
$\tau(u_2)$, $H_a(u_2)$, s_1 (quadrature)	0.1593550, 9.721560, 2.079588	0.1593550, 9.721560, 2.079588	6.4×10^{-16}	1×10^{-12}
s_2, $H_{a,\rm mid}$	-0.0009864616, 30.71391	-0.0009864616, 30.71391	1.6×10^{-11}	1×10^{-8}
maximal errors of the regime formulas	6.651898×10^{-6}, 0.0008521318, 0.0003210197, 0.1046619	6.651898×10^{-6}, 0.0008521318, 0.0003210197, 0.1046619	5.3×10^{-8}	0.001
crossover times $\tau(H_a = 10^3)$, $\tau(H_a = 1)$	0.001147118, 1.394714	0.001147118, 1.394714	4.6×10^{-13}	1×10^{-9}
coupled, near the singularity	0.7500003, -136.0829, 1224.744, 2.250007, -1.250007	0.7500003, -136.0829, 1224.744, 2.250007, -1.250007	1.4×10^{-13}	1×10^{-8}
coupled: ($p_{\rm eff}$, gas fraction) in the three Lovelock regimes	0.9993183, 0.001188768, 1.559665, 1.845034×10^{-7}, 0.5470861, 2.832020×10^{-5}	0.9993183, 0.001188768, 1.559665, 1.845034×10^{-7}, 0.5470861, 2.832020×10^{-5}	6.0×10^{-12}	1×10^{-6}
coupled, gas-dominated late regime	0.0003148578, 0.3999953, 1.873114×10^{-6}, 0.2499986	0.0003149595, 0.3999952, 1.873596×10^{-6}, 0.2499986	0.00032	0.002
growth exponent of the gas fraction (Einstein regime)	1.169833, 1.160189	1.169833, 1.160189	1.5×10^{-11}	1×10^{-8}
pair creation $F(1), F(8), F(16)$ and the tail exponent	0.01074935, 7.104066×10^{-6}, 8.157299×10^{-7}, -3.122482	0.01074935, 7.104066×10^{-6}, 8.157299×10^{-7}, -3.122482	1.2×10^{-9}	1×10^{-6}
brane plasma fraction at $T = H = M_8$	1.974364×10^{-28}	1.974364×10^{-28}	2.3×10^{-16}	1×10^{-12}
capture: start τ, β	56.75049, 0.4595957	56.75049, 0.4595957	3.8×10^{-12}	1×10^{-8}
capture: A, ϑ, τ_E, residual	0.4693305, 2.136009, -1564.358, 0.01278625	0.4693305, 2.136009, -1564.358, 0.01278625	3.9×10^{-9}	1×10^{-6}
capture: $\langle H_E \rangle(\tau - \tau_E)$	0.6669421, 0.6667362	0.6669421, 0.6667362	9.2×10^{-12}	1×10^{-8}
capture: gas share, radion share	0.008595793, 0.008728032, 0.9912720	0.008595793, 0.008728032, 0.9912720	1.5×10^{-10}	1×10^{-8}
radion decay: Γ, lifetime, $T_{\rm RH}$	4.572169×10^{-37}, 1.439606×10^{12}, 1.012615×10^{-9}	4.572169×10^{-37}, 1.439606×10^{12}, 1.012615×10^{-9}	3.7×10^{-16}	1×10^{-12}
Kaluza–Klein momenta below $m_r/2$	1.173352×10^{13}	1.173352×10^{13}	1.7×10^{-16}	1×10^{-12}
radion decay into the bulk: lattice sum, continuum limit, lifetime, ratio	5.999169×10^{-24}, 5.999169×10^{-24}, 0.1097172, 1.312106×10^{13}	5.999169×10^{-24}, 5.999169×10^{-24}, 0.1097172, 1.312106×10^{13}	8.6×10^{-16}	1×10^{-9}
$V_\perp$, $\ell_\perp$, $2\pi/\ell_\perp$	5.929225×10^{12}, 3.079186×10^{-13}, 0.004026525	5.929225×10^{12}, 3.079186×10^{-13}, 0.004026525	2.2×10^{-16}	1×10^{-12}
normalcy temperature T_*	13.26478	13.26478	0	1×10^{-12}
$g_*(T)$, $g_{*s}(T)$	106.7941, 106.7785, 86.29629, 86.20451, 76.41793, 75.57706 …	106.7941, 106.7785, 86.29629, 86.20451, 76.41793, 75.57706 …	5.5×10^{-9}	1×10^{-6}
$t(T)$	2.794096×10^{-7}, 6.123537×10^{-5}, 0.007393003, 0.7431868, 118.8395, 13161.66	2.794096×10^{-7}, 6.123537×10^{-5}, 0.007393003, 0.7431868, 118.8395, 13161.66	2.7×10^{-9}	1×10^{-6}

quantity	7+1.nb	Python	max. relative difference	tolerance
$1/(2H_a)$ divided by $t(T)$	0.9907571, 0.9234161, 0.9943226, 0.9985476, 0.9789280, 0.9996414	0.9907571, 0.9234161, 0.9943226, 0.9985476, 0.9789280, 0.9996414	7.6×10^{-11}	1×10^{-6}
Ω_r, Ω_Λ, $z_{\rm eq}$, age	9.209579×10^{-5}, 0.6849079, 3419.352, 13.79069	9.209579×10^{-5}, 0.6849079, 3419.352, 13.79069	1.7×10^{-15}	1×10^{-9}
maximal error of the composite late-time solution	0.0002201573	0.0002201573	6.6×10^{-12}	1×10^{-6}
radion displacement today	1.741296×10^{-86}, 5.621870×10^{-24}	1.741296×10^{-86}, 5.621870×10^{-24}	4.1×10^{-16}	1×10^{-12}
$\alpha_2 H_0^2$, $\alpha_3 H_0^4$	2.067046×10^{-92}, $4.272681 \times 10^{-190}$	2.067046×10^{-92}, $4.272681 \times 10^{-190}$	1.0×10^{-15}	1×10^{-12}
H_∞, Gibbons–Hawking temperature	1.807696×10^{-18}, 1.189847×10^{-42}, 2.197548×10^{-30}	1.807696×10^{-18}, 1.189847×10^{-42}, 2.197548×10^{-30}	3.2×10^{-16}	1×10^{-12}
$2\pi M/H_\infty$	5.280666×10^{33}, 5.280666×10^{42}	5.280666×10^{33}, 5.280666×10^{42}	2.3×10^{-16}	1×10^{-12}
de Sitter radion test	0.01020408, 0.01010359, 0.01020408, 0.01010359	0.01020408, 0.01010359, 0.01020408, 0.01010359	8.5×10^{-16}	1×10^{-9}
shares of the four terms of the 00 equation (section 9)	5.730063×10^{-10}, 0.5011390, 0.4976723, 0.001188768, 0.0002353490, 0.9988001 …	5.730063×10^{-10}, 0.5011390, 0.4976723, 0.001188768, 0.0002353490, 0.9988001 …	1.0×10^{-11}	1×10^{-6}

The largest relative difference is 0.00032. The checks of sections 20–23 of the notebook:

#	check	result	section
1	cubic Lovelock density in invariant form: L_3 == R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f R_e^a_f^b (exact, four random integer algebraic curvature tensors)	True	20.1
2	Gauss-Bonnet density in invariant form: L_2 == R^2 - 4 Ric.Ric + Riem.Riem (the same tensors, exact)	True	20.1
3	stabilizing potential: W = a^3 b^4 U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b)	True	20.2
4	brane tension smeared over the torus (V_perp = b^4 Lc^4): eps = f_W/V_perp, pa = -f_W/V_perp, pb = 0	True	20.2
5	brane fluid rho_4 = N a^(-3(1+w)), smeared: eps = rho_4/V_perp, pa = w eps, pb = 0	True	20.2
6	massless dirac16complex particle (comoving momentum (k_a, k_b)): eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega)	True	20.2
7	every such source is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 for arbitrary W(a, b), a(x0), b(x0)	True	20.2
8	minisuperspace: EL_n == (a^3 b^4/kappa)(S00 - kappa eps), EL_a == (3 a^2 b^4/kappa)(S11 + kappa pa), EL_b == (4 a^3 b^3/kappa)(S44 + kappa pb) with S = Sum alpha_k E^(k) (identities of section 12.6)	True	20.2
9	radion equation of Einstein gravity: Hb' + 3 Ha Hb + 4 Hb^2 == kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (constraint used)	True	20.3
10	radion mass: U = U0 + (K_U/2) beta^2, b U' = K_U beta gives the restoring term -(kappa K_U/12) beta, m_r^2 = kappa K_U/12	True	20.3
11	static auxiliary space: (kappa/4) b U' == Lambda + kappa U + 2 kappa f_W/V_perp + (kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2)	True	20.3
12	static auxiliary space: Ha' == -(kappa/2)(eps + pa) and E^(3) does not contribute (Hb = 0)	True	20.3
13	brane Friedmann equation: 3 Ha^2 == (kappa/V_perp) rho_4 + Lambda4 with Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp, i.e. 8 Pi G4 = kappa/V_perp	True	20.3
14	Einstein-frame form of the constraint: 3 (Ha + 2 Hb)^2 == 6 Hb^2 + (E^(1)0_0 of section 12.6), i.e. 3 M_Pl^2 H_E^2 == phi'^2/2 + rho with phi = Sqrt[12] M_Pl beta	True	20.3
15	vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): P_E + X P_G + xi X^2 P_C == 0 with P_E = 3(2u^2 + 4u + 1), P_G = -12u(u^3 + 12u^2 + 18u + 4), P_C = 360 u^3 (3u + 4)	True	21.1
16	Kasner constants: P_E(u1) == 0, p1 == (1 + 2 Sqrt[2])/7, q1 == (2 - 3 Sqrt[2])/14; u2 in (-1/3, 0) with P_G(u2) == 0, 3 p2 + 4 q2 == 3	True	21.1
17	the explicit curve solves the vacuum constraint: residual < 10^-25 (relative) at u = u1 + 10^-6, -0.28, u2, -0.1, -10^-4 (xi = 10^-6, 40 digits)	True	21.1
18	the effective exponent along the curve: p_eff -> p1 (u -> u1), about p2 at u = u2 (within 10^-2), -> 1 (u -> 0)	True	21.1

#	check	result	section
19	cubic regime: Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves the 00, 11, 44 equations through order tau^-1 (11: checked, not used)	True	21.2
20	cubic regime coefficients: h1 == alpha2/(16 alpha3) - 1/(32 alpha2), A0 == Sqrt[30](58 alpha2^2 - 45 alpha3)/(1920 alpha2^(3/2) alpha3^(1/2)), A1 and h2 as in the text	True	21.2
21	Einstein regime: u = u1 + d1 Ha^2, d1 == alpha2 (16 - 11 Sqrt[2])/4; p_eff = p1 + e1 Ha^2, e1 == alpha2 (1 + 9 Sqrt[2])/7	True	21.2
22	Gauss-Bonnet regime: p_eff = p2 + f1/Ha^2 + f2 xi Ha^2 with f1 = -0.69887346..., f2 = -0.7853407... (differences give the same values at Ha = 10^4 and 10^-6 xi Ha^2 to 10^-5)	True	21.2
23	cubic-series data at tau = 10^-8 satisfy the vacuum constraint to < 10^-12 (relative to 48 Ha^3 |Hb|)	True	21.2
24	the ODE solution agrees with the exact quadrature: tau at Ha(u2) to 10^-10, and tau - p1/Ha + e1 Ha at Ha = 3 10^-3 equals s1 to 10^-6	True	21.2
25	regime approximations: cubic error < 10^-5 for Ha >= 10^4, Gauss-Bonnet < 10^-3 for 3 <= Ha <= 300, Einstein < 10^-3 for Ha <= 0.1	True	21.2
26	matter-dominated cubic regime: p_eff -> 3/4, Hb -> -Sqrt[alpha2/(54 alpha3)], kappa eps tau^3 -> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (Ha = 10^15)	True	21.3
27	coupled: where the gas is a small fraction the effective exponents are those of the vacuum (1, p2, p1 within 10^-2)	True	21.3
28	without the stabilizing potential the gas-dominated Einstein regime decompactifies: tau Ha -> 0, tau Hb -> 2/5, pb/eps -> 1/4 (Ha = 10^-14)	True	21.3
29	the gas fraction grows like tau^(2 - 3 p1 - 5 q1) = tau^1.160 in the Einstein regime (to 2 10^-2)	True	21.3
30	Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (0, 2/5); (1/3, 0) -> (1/2, 0); (1/7, 1/7) -> (1/4, 1/4) (tau^2 E^(1) from EKas of section 14.1)	True	21.3
31	matter-dominated cubic regime (any fluid, b finite): leading 44 term p^3 (p - 1)(648 alpha3 h^2 - 12 alpha2); with h^2 = alpha2/(54 alpha3): p = 1/(1 + w_a) and kappa eps tau^3 = (64/3) p^3 alpha2 |h|	True	21.3
32	Einstein-Kasner pair creation: F(1) == 0.0107493 (section 14.3) and the tail F ~ K^(-(3 Sqrt[2] + 2)/2) (exponent between K = 8 and 16 to 2 10^-2)	True	21.4
33	brane plasma at T = M8, H = M8: fraction (Pi^2 g*/30) M8^4/(3 M_Pl^2 M8^2) < 10^-27	True	21.4
34	capture: the radion oscillates like a four-dimensional massive field in a matter-dominated universe: with tau_E from <H_E> = 2/(3 (tau - tau_E)) over the last period, beta = A sin(m_r tau + vartheta)/(m_r (tau - tau_E)) to 2 10^-2, <H_E>(tau - tau_E) = 2/3 to 10^-2 at tau = 10^5 and 2 10^5, the gas share constant to 2 10^-2	True	22.1
35	through brane fields alone a 10 GeV radion would decay far too late: T_RH < 10^-6 GeV (nucleosynthesis needs 4 10^-3 GeV); 10^13 Kaluza-Klein channels below m_r/2	True	22.2
36	the number of lattice vectors with |n|^2 = k is r_4(k) = 8 sigma(k) - 32 sigma(k/4) (direct count for k <= 60)	True	22.2
37	the radion decays into the bulk: the lattice sum over the Kaluza-Klein modes of dirac16complex (m << m_r/2) equals its continuum limit to 10^-6, the number of momenta to 10^-5, and Gamma_bulk/Gamma_brane > 10^12	True	22.2
38	g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), 2 + (21/4)(4/11)^(4/3)(3.044/3) = 3.3855 after e+e- annihilation (to 10^-3)	True	22.3
39	the time-temperature quadrature is converged: halving the grid changes t(10 keV) by less than 10^-5	True	22.3
40	radiation era: t(1 MeV) = 0.74 s within 5% (1/(2 Ha) with g* = 10.75)	True	22.3
41	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True	22.4
42	composite analytic solution t(a) = t_rm(a) + t_mLambda(a) - t_m(a) accurate to < 3 10^-4 for 10^-8 <= a <= 10^0.5	True	22.4
43	far future: Ha(t) = H_inf Coth[3 H_inf t/2] agrees with the full brane solution at a = 3 and a = 10 to 10^-4	True	23.1
44	radion in de Sitter: gamma/(3H) and m_eff^2/m_r^2 as in the text (beta_eq -> 0); gamma = 3H and m_eff^2 = m_r^2 for alpha2 = alpha3 = 0	True	23.2
45	radion equilibrium in de Sitter: 3 m_r^2 beta_eq == 3 H^2 - 12 alpha2 H^4 + O(beta_eq^2)	True	23.2
46	the radion relaxes to the equilibrium of the two algebraic equations: beta(3000) to 10^-6, Ha(3000) to 10^-8 (relative)	True	23.2
47	eight-dimensional de Sitter (no stabilizer): all three equations read 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 == Lambda	True	23.2
48	the shares of the four terms of the 00 equation: Einstein > 0.99 at Ha = 10^-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic about 1/2 each at Ha = 3 10^5, gas > 0.1 at Ha = 10^15	True	23.3

The Python program's boolean checks:

#	check	result
1	stabilizing potential: S_U = -Int sqrt|g| U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b)	True
2	brane tension, smeared over the torus (delta^4(y) -> 1/Lc^4): eps = f_W/V_perp, pa = -f_W/V_perp, pb = 0, V_perp = b^4 Lc^4	True
3	brane fluid (Standard Model plasma, rho_4 = N_W a^(-3(1+w))), smeared: eps = rho_4/V_perp, pa = w eps, pb = 0	True
4	massless dirac16complex particle (comoving momenta k_a, k_b): eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega), V = a^3 b^4	True
5	every such matter is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for arbitrary W(a, b), a(x0), b(x0)	True
6	field equations from S_grav + S_m: EL_n = (a^3 b^4/kappa)(S00 - kappa eps), EL_a = (3a^2 b^4/kappa)(S11 + kappa pa), EL_b = (4a^3 b^3/kappa)(S44 + kappa pb), S = Sum alpha_k E^(k) (minisuperspace, lapse n = 1)	True
7	radion equation (Einstein): Hb' + 3 Ha Hb + 4 Hb^2 = kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (Hb = beta', exact)	True
8	radion mass: with U = U0 + (K/2) ln^2(b/b0) the restoring term is -m_r^2 beta, m_r^2 = kappa K/12	True
9	static auxiliary space: (kappa/4) b U'(b) = Lambda + kappa U + 2 kappa f_W/V_perp + (kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2)	True
10	static auxiliary space: E^(3) does not contribute, and Ha' = -(kappa/2)(eps + pa) (brane Friedmann acceleration)	True
11	four-dimensional Friedmann equation on the brane (b static): 3 Ha^2 = 8 pi G4 rho4 + Lambda4 with 8 pi G4 = kappa/V_perp and Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp	True
12	cubic Lovelock density: L_3 = R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f R_e^a_f^b (exact, four random integer algebraic curvature tensors)	True
13	Gauss-Bonnet density: L_2 = R^2 - 4 Ric.Ric + Riem.Riem (same tensors)	True
14	vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): E(u) + X G(u) + xi X^2 K(u) = 0 with E = 3(2u^2 + 4u + 1), G = -12u(u^3 + 12u^2 + 18u + 4), K = 360 u^3 (3u + 4)	True
15	Einstein regime: u1 = Hb/Ha = -1 + 1/sqrt(2), root of E(u); Kasner p1 = 1/(3 + 4u1) = (1 + 2 sqrt 2)/7, q1 = u1 p1 = (2 - 3 sqrt 2)/14	True
16	Gauss-Bonnet regime: u2 = Hb/Ha, root of u^3 + 12u^2 + 18u + 4 in (-1/3, 0); Kasner p2 = 3/(3 + 4u2), q2 = u2 p2 (3p + 4q = 3)	True
17	exact vacuum solution: for every u in (u1, 0) the curve Ha = sqrt(X(u)), Hb = u Ha, X(u) = [G + sqrt(G^2 - 4 xi K E)]/(2 xi |K|), solves the constraint (residual < 1e-20 relative)	True
18	the effective exponent goes from p1 (Einstein, u -> u1) through about p2 (Gauss-Bonnet, u = u2) to 1 (cubic, u -> 0)	True
19	cubic regime (tau -> 0): Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves all three equations to the orders used; h0 = -sqrt(alpha2/(30 alpha3))	True
20	cubic regime: h1 = alpha2/(16 alpha3) - 1/(32 alpha2), A0 = sqrt(30)(58 alpha2^2 - 45 alpha3)/(1920 alpha2^(3/2) alpha3^(1/2))	True
21	cubic regime: A1 = -3647 alpha2/(92160 alpha3) - 223/(6144 alpha2) + 645 alpha3/(8192 alpha2^3), h2 = sqrt(30)(-1212 alpha2^4 + 1180 alpha2^2 alpha3 + 225 alpha3^2)/(307200 alpha2^(5/2) alpha3^(3/2))	True
22	Einstein regime (Ha -> 0): u = u1 + d1 Ha^2 + ..., d1 = alpha2 (16 - 11 sqrt 2)/4; p_eff = p1 + e1 Ha^2 + ..., e1 = alpha2 (1 + 9 sqrt 2)/7	True
23	cubic-series initial data at tau = 1e-8 satisfy the constraint to < 1e-12 (relative to 48 alpha2 Ha^3 |Hb|)	True
24	the ODE solution and the quadrature agree: tau at Ha(u2) to 1e-10, s1 (ODE at Ha = 3e-3) to 1e-6	True
25	regime approximations: cubic error < 1e-5 for Ha >= 1e4, Gauss-Bonnet < 1e-3 for 3 <= Ha <= 300, Einstein < 1e-3 for Ha <= 0.1	True
26	matter-dominated cubic regime: p_eff -> 3/4, Hb -> -sqrt(alpha2/(54 alpha3)), kappa eps tau^3 -> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (to 1e-3)	True
27	shares: Einstein > 0.99 at Ha = 1e-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic about 1/2 each at Ha = 3e5 (to 1e-2), gas > 0.1 at Ha = 1e15	True
28	coupled: where the gas is a small fraction, the effective exponents are those of the vacuum solution (1, p2, p1 within 1e-2)	True
29	without a stabilizing potential the gas-dominated Einstein regime decompactifies: a -> const (tau Ha -> 0), b ~ tau^(2/5) (tau Hb -> 2/5), pb/eps -> 1/4	True
30	gas fraction grows like tau^(2 - 3p1 - 5q1) = tau^1.160 in the Einstein regime (to 2e-2)	True
31	Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (P, Q) = (0, 2/5) [decompactification]; (1/3, 0) -> (1/2, 0) [brane radiation, b frozen]; (1/7, 1/7) -> (1/4, 1/4) [isotropic]	True
32	matter-dominated cubic regime (any fluid pa = w_a eps, b finite): leading 44 equation p^3 (p - 1)(648 alpha3 h^2 - 12 alpha2) = 0 -> h^2 = alpha2/(54 alpha3); 11/00 -> p = 1/(1 + w_a); kappa eps tau^3 = (64/3) p^3 alpha2 |h|	True

#	check	result
33	pair creation has a power-law tail F ~ K^(-2(p1 - q1)/(1 - p1)) = K^(-(3 sqrt 2 + 2)/2) = K^(-3.121) (to 2e-2): it comes from the singularity, so the number of created pairs is dominated by the earliest times	True
34	F(1) agrees with the value of section 14.3 of 7+1.nb (|beta|^2 for (|p|, |q|) = (1, 1))	True
35	the Standard Model plasma on the brane (T <= M8) is gravitationally negligible in the bulk while H ~ M8: fraction < 1e-27	True
36	capture: the radion oscillates as a four-dimensional massive field in a matter-dominated universe: j0 form to 2e-2, <H_E>(tau - tau_E) -> 2/3 to 1e-2, the gas fraction stays constant to 2e-2	True
37	four-dimensional (Einstein-frame) form of the 00 equation: 3 (Ha + 2Hb)^2 = 6 Hb^2 + kappa eps_total, i.e. 3 H_E^2 M_Pl^2 = (1/2) phi'^2 + rho with phi = sqrt(12) M_Pl ln(b/b0)	True
38	through brane fields alone a radion of 10 GeV would decay far too late (T_RH < 1 keV < 4 MeV needed by nucleosynthesis): the moduli problem of the minimal model	True
39	Jacobi's four-square formula r_4(k) = 8 sigma(k) - 32 sigma(k/4) equals the direct count of lattice vectors for 1 <= k <= 60	True
40	the radion decays into the bulk: lattice sum = continuum limit to 1e-6, number of momenta = N_KK to 1e-5, Gamma_bulk/Gamma_brane > 1e12	True
41	g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), and after e+e- annihilation 2 + (21/4)(4/11)^(4/3)(N_eff/3) = 3.3855 (to 1e-3)	True
42	radiation era: the t(T) quadrature is converged (halving the grid changes t(10 keV) by < 1e-5)	True
43	radiation era: t(1 MeV) = 0.74 s within 5% (standard value 1/(2H) with g* = 10.75)	True
44	age of the universe 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True
45	the composite analytic solution (exact radiation+matter, exact matter+Lambda, minus their common matter part) is accurate to < 3e-4 at all times	True
46	far future: H(t) = H_inf coth(3 H_inf t/2) (exact for matter + Lambda) agrees with the full solution at a = 3, 10 to 1e-4	True
47	far future: the radion obeys beta'' + gamma beta' + m_eff^2 beta = 0 around de Sitter (constraint used for delta Ha); alpha2 = alpha3 = 0: gamma = 3H, m_eff^2 = m_r^2 at beta_eq -> 0	True
48	far future: equilibrium of the radion in de Sitter: 3 m_r^2 beta_eq = 3H^2 - 12 alpha2 H^4 (+ O(beta_eq^2))	True
49	eight-dimensional de Sitter (no stabilizer): 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 = Lambda (all three equations)	True
50	far future: the radion relaxes to its shifted equilibrium (to 1e-6 relative) and Ha to H_eq (to 1e-8)	True

12. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated).
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_coupled.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.

13. References

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